PROJECTPROPOSAL

PROJECTPILOT AI DISCOVERY ENGINE

Scope Architecture & Estimate: banking app

Prepared For:	ali (u2024156@giki.edu.pk)
Business Domain:	FINANCE
Target Platforms:	MOBILE
Generation Date:	July 20, 2026

1. Executive Summary

The proposed project 'banking app' is designed to serve as a high-quality finance platform. Built for mobile, it incorporates core features including auth, payments, notifications, ai. The project aims to build a secure, audit-compliant financial application tailored for modern fintech services. Security and accuracy are the core pillars of the proposed solution. The system features multi-factor authentication, role-based access controls, and detailed ledger logging for all transactions. The backend architecture prioritizes data encryption at rest and in transit.

2. Project Objectives

The primary business objective is to architect and launch a scalable, secure, and production-quality software application. Through ProjectPilot AI's automated discovery session, we have analyzed requirements, selected platforms, and structured business modules to maximize engineering velocity while keeping operational expenditures low.

■ **Time-to-Market**	Accelerate launch cycles by using a modular, containerized stack (FastAPI & React).
■ **Compliance & Security**	Integrate native encryption, OAuth2 verification, and domain security mapped to finance standards.
■ **Future Expansion**	Implement clean API separation, allowing effortless third-party integrations and smart automation additions.

3. Scope of Work & Features

The total scope covers a customized combination of standard features selected by the client, combined with intelligent additions recommended by the AI Architect based on industry standards.

Core Selected Features

- ✔ **Auth**
- ✔ **Payments**
- ✔ **Notifications**
- ✔ **Ai**
- ✔ **Admin**
- ✔ **Upload**
- ✔ **Search**
- ✔ **Dashboard**
- ✔ **Analytics**
- ✔ **Export**
- ✔ **Thirdparty**

AI-Recommended Enhancements

Recommended Feature	Value-Add & Purpose	Effort
Advanced Analytics Module	Custom reporting and visual data monitoring widgets.	**Medium**

4. Solution Architecture

The platform is structured on a modern, decoupled architecture. Client interactions happen via a responsive Single Page Application (SPA), sending secure, rate-limited requests to a high-speed async backend.

Presentation Layer	Service Layer	Persistence Layer
Vite + React SPA Framer Motion UI Responsive Design	FastAPI Async server Uvicorn Router JSON Serialization	PostgreSQL relational DB SQLAlchemy Engine Redis Cache

Technology Stack Specifications

Infrastructure Layer	Recommended Technology Stack
Frontend	React Native (Expo, NativeWind) for mobile app deployment
Backend	FastAPI (Python) with high-security middleware, OAuth2 + JWT token authentication.
Database	PostgreSQL with strict constraints, schema migrations (Alembic), and write-ahead transaction logs.
Hosting & Infrastructure	AWS isolated VPC, SSL/TLS terminates at load balancer, CloudTrail logging.
External Integrations	Plaid API for bank account connections, Plivo for SMS-based 2FA.

5. Project Implementation Roadmap

Phase / Milestone	Deliverables Scope	Est. Duration
Phase 1: Foundations & User Core	• Database Schema setup • Core API configurations • User Accounts & Authentication module	**2 Weeks**
Phase 2: Product Core Modules	• Payments module development • Notifications module development • Ai module development • Admin module development • Upload module development • Search module development • Dashboard module development • Analytics module development • Export module development • Thirdparty module development	**3 Weeks**
Phase 3: QA, Security & Deployment	• Quality Assurance & Integration tests • Performance optimization runs • PCI audits & billing live runs check • Cloud production environment deployment	**1 Week**

Agile Sprint Planning

Sprint	Sprint Objectives & Effort	Deliverables Checklist
Sprint 1	*Objectives:* Establish infrastructure, build base APIs, and set up landing dashboard frames. **Effort:** 364 Hours	• Docker infrastructure and API templates initialized • Database migrations complete and connected locally • User account registration UI & Backend logic verified
Sprint 2	*Objectives:* Complete end-to-end integration tests, resolve UI issues, and deploy to staging. **Effort:** 364 Hours	• Full integration unit test suite checks passing • Responsive styling issues resolved across web/mobile viewports • Continuous Integration (CI/CD) pipelines running successfully

6. Team Allocation & Budget Breakdown

Based on project parameters, the scope of selected features, and platform requirements, we have computed target effort metrics and engineered a precise allocation layout.

Total Est. Effort	Est. Timeline Duration	Engineering Team Size
728 hours	**7-9 weeks**	**5 resources**

Allocated Roles: Project Manager, UI/UX Designer, Frontend Developer, Backend Developer, QA Engineer

Category Component	Scope Details	Estimated Cost (USD)
UI/UX Design Phase	User experience wireframes, interactive mockup builds	$10,920.00
Frontend Development	React / Mobile application interface construction	$25,480.00
Backend Engineering	FastAPI endpoints, ORM connections, databases setup	$21,840.00
Quality Assurance	End-to-end integration tests, responsiveness check	$8,736.00
Project Management	Agile coordination, milestones, developer logs tracking	$5,824.00
Hosting & Cloud Infrastructure	Initial VPS staging sandbox setup & pipeline actions	$500.00
TOTAL BUDGET	**Estimated total investment, excluding hosting**	**$72,800.00**

7. Technical Risks & Mitigations

Identified Risk Factor	Proposed Mitigation Strategy
Compliance issues and regulatory audits	Ensure database logging stores all system modifications; maintain detailed audit trails.
Data leakage of sensitive customer financial records	Implement AES-256 field-level encryption for personal identifiable information (PII).
Distributed Denial of Service (DDoS) on transaction API	Set up AWS Shield, Cloudflare rate limiting, and backend request throttling.

8. Project Execution Next Steps

To initiate the development workflow, we recommend the following phases:

1. Discovery Alignment: Conduct a developer kick-off session to align architectural preferences and integrations.
2. Environments Initialization: Provision SQLite/PostgreSQL staging databases and hook continuous integration (CI) actions.
3. Milestone Sprint 1: Execute Sprint 1 scope, starting with authentication configurations and dashboard scaffolding.